

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460663.927 +/- 0.026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.042
Transit depth (Rp/Rs)^2	2.51 +/- 1.34 [%]
Orbital Inclination (inc)	85.07 +/- 2.46
Airmass coefficient 1 (a1)	1.0396 +/- 0.0089
Airmass coefficient 2 (a2)	-0.029 +/- 0.096
Scatter in the residuals of the lightcurve fit is	0.86 %
Best Comparison Star	None
Optimal Aperture	4.05
Optimal Annulus	10.53
Transit Duration (day)	0.06 +/- 0.037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.158 ± 0.042 vs 0.1276 (deviation 0.72σ)
Duration fit vs published	0.06 d vs 0.1179 d (deviation 1.57σ)
Mid-time O–C	16.8 min
Depth SNR	3.8
Residual scatter	0.86 %

Light curve

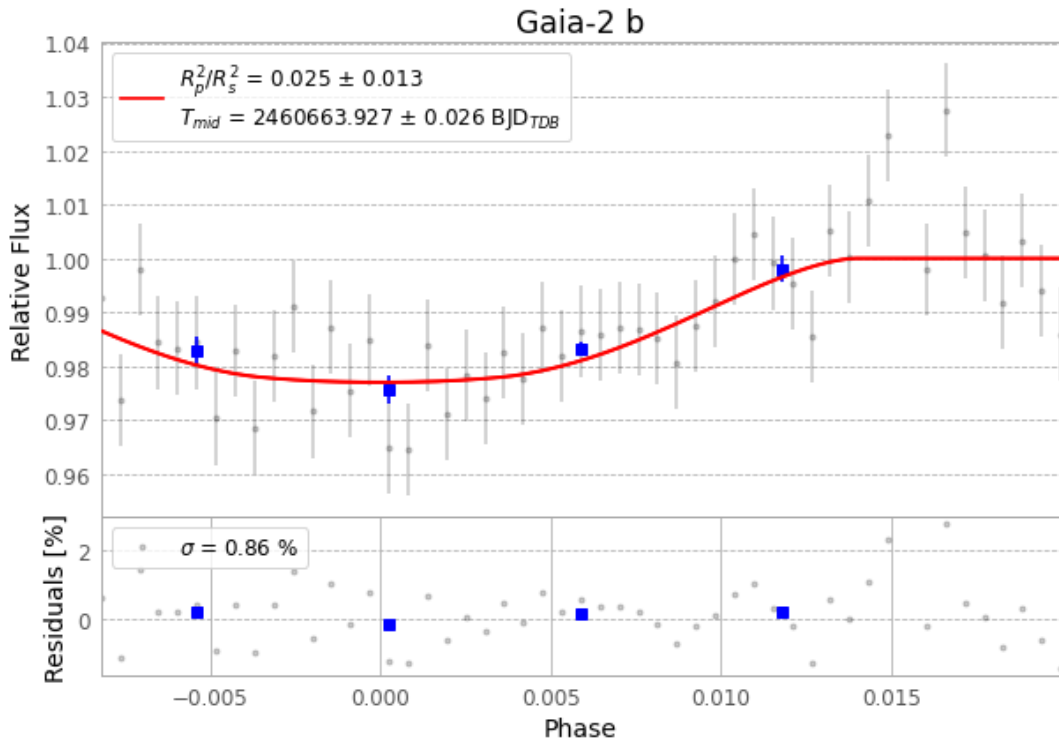


Figure 1 — FinalLightCurve_Gaia-2 b_2024-12-19.png (SHA-256 ee2ab086b03baa07...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [401, 289], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 354c04629f852b1a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

51 FITS frames (33 MB), Gaia-2241219092416.FITS → Gaia-2241219115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Gaia-2-241219.log (5d186f5f4085...), FinalLightCurve_Gaia-2 b_2024-12-19.png (ee2ab086b03b...), FinalLightCurve_Gaia-2 b_2024-12-19.csv (99463c4fa6aa...)

Compute resources

Wall time 82.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 23:05Z.